After Hours and Special Materials

Building access, dry ice, liquid nitrogen, and what to do when a freezer alarms at night

Version 2, 9 March 2026. Tomasz Brenner (Lab Safety Officer) and Marisol Vega (Lab Manager). Replaces the 2024 “Night owl” sheet.

1. Getting into the building after hours

- Keller Hall’s outside doors are locked from 7 pm to 7 am on weekdays, and all day on weekends and university holidays.
- Your Westbrook ID card opens the **east entrance** and the fourth-floor lab corridor after hours only if Marisol has requested after-hours access for you. She requests it after you complete Laboratory Safety Fundamentals, Hazardous Chemical Waste Management and Tomasz’s lab safety orientation (see *Welcome and First Week*).
- Do not prop doors open, and do not let people you do not know follow you in. If someone makes you uneasy, call University Police (555-0142).
- The campus SafeWalk escort (555-0144) will walk you to other campus buildings or bus stops at night.
- **After 8 pm, post your arrival and departure in #after-hours** on the lab Slack, with a line on what you are doing. This is a check-in, not a buddy. The buddy rules for hazardous and BSL-2 work are in *Health and Safety*, Section 7, and still apply.

2. Dry ice

Where to get it

- **Weekdays, until 5 pm:** the MCB stockroom, Keller B12, sold by the kilogram and charged to the lab account. Bring a foam cooler.
- **After 5 pm, at weekends and on holidays:** the **shared dry-ice chest in the loading dock of Pemberton Hall** (the chemistry building; the dock is on the north side, door P-2).
  - *Card access:* your ID card needs “Pemberton Dock” access. Marisol requests it through Facilities; allow about a week. It is only granted after you have completed Cryogen Safety training.
  - *Log sheet:* every time you take dry ice, write the date, time, your name, “Okafor Lab” and the kilograms taken on the clipboard sheet on the chest. The lab is billed monthly from this sheet.
  - *How much:* **at most 5 kg per person per visit**, and only what you need that night. If the chest is less than a quarter full, take at most 2 kg and email Marisol so that other labs are not left without dry ice in an emergency. The stockroom restocks the chest on Monday, Wednesday and Friday mornings.
- **Shipping on dry ice** is a regulated shipment: Marisol prepares all of them. Do not pack or send one yourself.

Handling dry ice safely

Dry ice is solid carbon dioxide at about -78 °C. It burns skin on contact, and as it sublimates it releases CO_2 gas, which can displace oxygen in enclosed spaces.

- Wear **insulated gloves** (cryogenic or thick insulated gloves) and safety glasses. Use a scoop, never your bare hands.
- Use and store it **only in well-ventilated areas**. **Never take dry ice into the walk-in cold room (416A)** or any other small, closed or poorly ventilated room.
- **Never put dry ice in a sealed container**: no screw-cap bottles, sealed jars, tightly taped boxes, or closed coolers with gasketed lids. The gas pressure can make them burst. Carry it in a foam cooler with a loose lid.
- Do not ride in a passenger lift with dry ice or cryogen dewars. Carry a small cooler up the stairs, or send the cooler up in the lift on its own with the "Cryogenic material, do not enter" sign from the 416 door, and have a colleague meet it.
- Do not put dry ice inside working freezers or refrigerators; it can damage the thermostat.
- To get rid of leftover dry ice, leave it to sublimate in the 412 fume hood with the sash lowered. **Never** put it down a sink or drain, or in a waste bin.

3. Liquid nitrogen (LN_2)

- **Who may dispense:** only people who have completed Cryogen Safety (online) *and* Tomasz's hands-on sign-off.
- **Supply:** the 180-L supply dewar in 416, delivered by Airgas on Monday mornings (Marisol orders it). The storage dewars (1 and 2) are topped up on Tuesdays and Fridays by the person on LN_2 duty (see the rota in *Who Does What*).
- **PPE:** face shield over safety glasses, loose-fitting cryogenic gloves (so you can shake them off quickly), lab coat, long trousers, closed shoes.
- Fill small (1–4 L) dewars at the supply dewar, using the withdrawal valve and phase separator. Only use approved dewars, and never put a tight lid on LN_2.
- **Buddy rule:** dispensing from the supply dewar after 8 pm, or at weekends, needs a second person present in the suite.
- **Oxygen alarm:** 416 has an oxygen monitor. If it alarms, **leave immediately**, close the door and do not go back in. If someone has collapsed inside, **do not go in after them**: call 911, then Tomasz.
- Do not ride in a lift with a dewar; send it up on its own with the sign, as for dry ice.
- Tell Marisol when the supply dewar reads below 30%.

4. When a -80 °C freezer alarms at night

The -80 freezers in 416 have audible alarms and remote monitoring. The monitoring system texts Marisol and Tomasz automatically when any freezer rises above -65 °C. If you are in the building and hear an alarm:

Step 0: your safety first. If you smell burning or see smoke, leave, pull the fire alarm, and call 911. Do not touch the electrical panel or reset breakers.
Step 1: check the freezer. Read the display. Is the door ajar? Close it firmly. Is the plug in? If after closing the door the temperature is below -70 °C and falling, silence the alarm, write a note on the freezer log sheet on the door, post in #after-hours, and you are done.
Step 2: call tree. If the freezer is above -70 °C and not recovering after 10 minutes, or it is off or making unusual noises, phone these people **in order** until you reach someone. Leave a voicemail *and* a text, and move to the next name after 5 minutes:

1. Marisol Vega, Lab Manager: mobile 555-0123
2. Tomasz Brenner, Lab Safety Officer: mobile 555-0127
3. Prof. Adaeze Okafor, PI: mobile 555-0110
4. Dr. Priya Raghunathan, postdoc: mobile 555-0129

If the whole floor has lost power, also call Facilities' 24-hour line: 555-0150.
Step 3: if nobody answers within 15 minutes and the freezer is above -60 °C: start moving its contents into **Freezer C** (the lab's backup, kept half-empty for this). If Freezer C is full or also failing, use the **department emergency freezer in Keller B20**; the key is in the lockbox by the 414 door and the code is pinned in #after-hours. Wear insulated gloves. Photograph each shelf before you move it, move whole racks and keep them in order, and write down what went where.
Step 4: afterwards. Record everything on the freezer log and email Marisol and Tomasz a short summary (which freezer, the times, the temperatures, what you moved).

Do not open the failing freezer's door repeatedly while you wait, and never try to repair it yourself.

CO_2 incubator "CO_2 low" alarm (413). The incubators switch automatically between two CO_2 cylinders. If the alarm sounds, check both cylinder gauges in the 413 gas cabinet. If both are empty, text Mei-Ling Zhou (mobile 555-0134) and Marisol. Do not change cylinders unless you have been trained to.

5. Before you leave at night

Check that the bench gas taps are closed. Wipe down the biosafety cabinets and close their sashes. Check that the ultracentrifuge and FPLC runs you booked are behaving. Close the freezer and cold-room doors firmly, and turn the lights off in empty rooms. Post your departure in #after-hours.